No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

CS6002 Project Report
Based on Balcan, Blum, Haghtalab, Procaccia (EC 2015)

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Abstract

We extend the online learning framework for repeated Stackelberg Security Games (SSGs) of Balcan, Blum, Haghtalab, and Procaccia [1] to the setting where the set of attacker types K is *not known* to the defender in advance, but its cardinality is bounded: $|K| \leq K_{\max}$. We assume full-information feedback. We present Grow-Catalogue, an algorithm that incrementally discovers new attacker types and maintains the set of approximate extreme points of the best-response polytopes. We prove that Grow-Catalogue achieves expected regret

$$\mathbb{E}[\mathrm{Regret}(T)] \;=\; O\Big(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\Big) \;=\; \tilde{O}\Big(n\sqrt{K_{\max}\, T}\Big),$$

matching the bound of Balcan et al. (their Theorem 5.1) with k replaced by $K_{\max}$. When $K_{\max} = O(1)$, our bound becomes $\tilde{O}(n\sqrt{T})$. We also characterize the parameter regime in which the bound is meaningful and we empirically validate the algorithm on synthetic SSG instances, comparing *total recomputation* and *incremental* variants of the polytope update, against a *known-K* oracle and a *uniform random* baseline.

1 Introduction

Balcan, Blum, Haghtalab, and Procaccia [1] studied the problem of designing no-regret algorithms for a defender in a repeated Stackelberg Security Game when the sequence of attackers is chosen adversarially from a *known* set $K = \{\alpha_1, \ldots, \alpha_k\}$ of attacker types. They proved a full-information regret bound of $O(\sqrt{Tn^2k\log(nk)})$ and a matching lower bound that is linear in T when the set of attacker types is unrestricted.

In many deployed security applications, the defender does *not* actually know in advance the full set of attacker types that may arrive. However, a natural structural assumption, which we adopt, is that the total number of distinct attacker types is bounded by some constant $K_{\max}$. Under this assumption we ask:

> *Can the defender achieve sublinear regret even when the set K is unknown, provided $|K| \leq K_{\max}$?*

We answer this question affirmatively. Our main contribution is an algorithm, Grow-Catalogue, that incrementally discovers attacker types, recomputes the relevant best-response polytopes and their approximate extreme points, and runs a multiplicative-weights style sub-routine over the growing expert set. We prove the algorithm achieves essentially the same regret bound as Balcan et al., with k replaced by the known bound $K_{\max}$. The bound degrades gracefully as $K_{\max}$ grows with T, and we characterize the precise regime in which the bound remains sublinear.

All assumptions we make beyond [1] are stated explicitly. The utilities, action set, and adversary model are identical to Balcan et al.; the *only* change is that the set K is unknown a priori (but $K_{\max}$ is known), and new types reveal their full utility functions upon first appearance (consistent with full-information feedback).

2 Preliminaries and Notation

We follow Balcan et al. [1, Section 2] exactly. A repeated Stackelberg Security Game is a tuple with the following components.

Game primitives.

- Time horizon $T \in \mathbb{N}$ (known to the defender).
- Set of targets $N = \{1, \ldots, n\}$.
- Defender resources R, schedules $\mathcal{D} \subseteq 2^N$, assignment function $A : R \to 2^{\mathcal{D}}$.
- The set of pure strategies is determined by $N, \mathcal{D}, R, A$. Let there be m pure strategies and let M be the zero–one $n \times m$ matrix with $M_{i,j} = 1$ iff target i is covered by some resource in the j-th pure strategy.
- A mixed strategy is a vector $\mathbf{s} \in \Delta_m$ (a distribution over pure strategies). Its induced *coverage probability vector* is $\mathbf{p} = M\mathbf{s} \in [0,1]^n$. Let $\mathcal{P} \subseteq [0,1]^n$ denote the set of all valid coverage probability vectors; $\mathcal{P}$ is a convex polytope.
- Defender utilities $u_d^c(i), u_d^u(i) \in [-1, 1]$ for each $i \in N$, giving $U_d(i, \mathbf{p}) = u_d^c(i)\, p_i + u_d^u(i)\,(1 - p_i)$, which is *linear* in $\mathbf{p}$ for each fixed i.

Attacker types. A type α is specified by utilities $u_\alpha^c(i), u_\alpha^u(i) \in [-1, 1]$ for each $i \in N$, yielding expected utility $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i)p_i + u_\alpha^u(i)(1 - p_i)$. The attacker of type α best-responds by playing

$$b_\alpha(\mathbf{p}) \;=\; \arg\max_{i \in N} U_\alpha(i, \mathbf{p}),$$

with an arbitrary but consistent tie-breaking rule.

Adversarial sequence. There is an (unknown) set K of attacker types. An *oblivious* adversary selects a sequence $\mathbf{a} = a_1, \ldots, a_T \in K^T$ before the game starts. At each round t the defender commits to a coverage probability vector $\mathbf{p}_t$, the attacker of type a_t best-responds, and the defender observes full feedback a_t (i.e., the identity of the attacker type).

Regret. The benchmark is the best fixed coverage in hindsight:

$$\mathbf{p}^* \;=\; \arg\max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}).$$

The expected regret of an online algorithm is

$$\mathrm{Regret}(T) \;=\; \sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; \mathbb{E}\left[\sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right],$$

where the expectation is over the algorithm's internal randomization.

Two structural objects from [1]. For a set of types $C \subseteq K$ and type $\alpha \in C$, target $i \in N$, define

$$\mathcal{P}_i^\alpha(C) \;=\; \{\mathbf{p} \in \mathcal{P} : b_\alpha(\mathbf{p}) = i\},$$

which is a convex polytope (it is defined by $n-1$ linear inequalities $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$ plus the membership constraints of $\mathcal{P}$). For a response assignment $\sigma : C \to N$, define

$$\mathcal{P}_\sigma(C) \;=\; \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha(C),$$

which is also a convex polytope (possibly empty). Let $\Sigma(C) = \{\sigma : \mathcal{P}_\sigma(C) \neq \emptyset\}$; the collection $\{\mathcal{P}_\sigma(C) : \sigma \in \Sigma(C)\}$ partitions $\mathcal{P}$ into convex regions in which every attacker type in C has a fixed best response.

We recall the approximation lemma of Balcan et al. [1, Lemmas 4.3–4.4], restated here for a general catalogue C.

Definition 1 (ϵ-approximate extreme points, [1] adapted)**.** *For $\epsilon > 0$ and $C \subseteq K$, let $\mathcal{E}(C; \epsilon)$ be a set of coverage vectors in $\mathcal{P}$ such that for every $\sigma \in \Sigma(C)$ and every exact extreme point $\mathbf{p}$ of the closure of $\mathcal{P}_\sigma(C)$: either $\mathbf{p} \in \mathcal{E}(C; \epsilon) \cap \mathcal{P}_\sigma(C)$, or there exists $\mathbf{p}' \in \mathcal{E}(C; \epsilon) \cap \mathcal{P}_\sigma(C)$ with $\|\mathbf{p} - \mathbf{p}'\|_1 \leq \epsilon$.*

Lemma 1 (Approximation by extreme points, [1] Lemma 4.3 restated)**.** *Let $C \subseteq K$ and let $\mathcal{E} = \mathcal{E}(C; \epsilon)$ as in Definition 1. For any sequence $\mathbf{a}' = a'_1, \ldots, a'_{T'}$ with $a'_t \in C$ for all t,*

$$\max_{\mathbf{p} \in \mathcal{E}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \;\geq\; \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \;-\; 2\epsilon T'.$$

Lemma 2 (Size of $\mathcal{E}(C; \epsilon)$, [1] Lemma 4.4 restated)**.** *For $\epsilon > 0$ small enough that the construction of Definition 1 is well-defined, there exists $\mathcal{E}(C; \epsilon)$ with*

$$|\mathcal{E}(C; \epsilon)| \;\in\; O\Big((2^n + |C| \cdot n^2)^n \cdot n^{|C|}\Big).$$

Proposition 1 (Black-box regret, [1] Proposition 1)**.** *There is an algorithm (e.g., Hedge with a time-varying learning rate [2]) that, for any fixed expert set $\mathcal{M}$ and any adversarial loss sequence with losses in $[-\kappa, \kappa]$, achieves expected regret*

$$R_{T', \mathcal{M}, \kappa} \;\leq\; O\Big(\sqrt{T' \cdot \kappa \cdot \log |\mathcal{M}|}\Big)$$

after any number T' of rounds, without requiring T' to be known a priori.

Remark 1. *Balcan et al. state Proposition 1 assuming T' is known. The extension to unknown T' is standard; see Cesa-Bianchi and Lugosi [2, Chapter 2], using either the doubling trick or a time-varying learning rate $\eta_t = \sqrt{\log |\mathcal{M}|/t}$. We need this extension because the length of an "epoch" in our algorithm is not known in advance.*

3 Problem Formulation: Unknown Set of Attacker Types

We now describe our setting, which differs from [1] in exactly one point.

Definition 2 (Unknown-K repeated SSG with bound $K_{\max}$)**.** *A repeated SSG with an unknown, bounded set of attacker types is identical to the setting of Section 2 except that:*

1. *The set K of attacker types is* unknown *to the defender at the start of the game.*

2. *The defender is given a constant $K_{\max} \geq 1$ such that $|K| \leq K_{\max}$.*

3. *(Full information, as in [1] Section 5.) At each round t, after committing $\mathbf{p}_t$, the defender observes a_t, i.e., the identity of the attacker type. The* first *time a type α is observed, the defender additionally receives its full utility function (u^c_α, u^u_α). (This is consistent with the full-information assumption: since the defender sees the attacker's identity, the same identity on subsequent rounds refers to the same fixed utility function.)*

All utilities take values in $[-1, 1]$, exactly as in [1, Section 2].

Remark 2 (Why $K_{\max}$ must be finite)**.** *Balcan et al. [1, Theorem 7.1] prove a lower bound: for any horizon T, there is a game with $|K| < 2^{T+1}$ in which any online algorithm incurs regret $\Omega(T)$. Hence* some *restriction on $|K|$ is necessary for sublinear regret. We assume the least restrictive such assumption: $|K| \leq K_{\max}$ for a known constant $K_{\max}$.*

Remark 3 (Benchmark unchanged)**.** *The benchmark $\mathbf{p}^*$ is defined with respect to the sequence $\mathbf{a}$ actually realized, which contains at most $K_{\max}$ distinct types. Since the defender knows $K_{\max}$ but not K in advance, it is trivially true that $\mathbf{p}^* \in \mathcal{P}$, and the best-in-hindsight comparison is meaningful.*

4 The Grow-Catalogue Algorithm

4.1 Intuition

At any point in time, the defender has observed some *catalogue* $C_t \subseteq K$ of types. Since $|K| \leq K_{\max}$, $|C_t| \leq K_{\max}$ at all times. The algorithm maintains the set $\mathcal{E}(C_t; \epsilon)$ of approximate extreme points corresponding to the current catalogue, and runs a Hedge sub-routine over this set. When a new type α is observed for the first time:

1. α is added to the catalogue.

2. The partition $\{\mathcal{P}_\sigma(C_t) : \sigma \in \Sigma(C_t)\}$ is refined: each existing region is split by the $n-1$ linear inequalities defining α's best responses. New extreme points are introduced.

3. A fresh instance of Hedge is started on the new expert set (this is the "total recomputation" variant). Alternatively, Hedge weights can be transferred for surviving experts (the "incremental" variant); the regret bound is identical.

Intuitively, the analysis partitions $[1, T]$ into epochs, one per distinct discovered type. Within each epoch the catalogue is fixed and we inherit Balcan et al.'s full-information analysis (Theorem 5.1). The only extra cost comes from (i) summing the per-epoch regrets across $K' \leq K_{\max}$ epochs, handled by a Cauchy–Schwarz step, and (ii) a constant per-epoch discovery-round cost, absorbed into the $O(K_{\max})$ lower-order term.

4.2 Pseudocode

Algorithm 1 GROW-CATALOGUE (full information)

Require: Time horizon T; bound $K_{\max}$; defender game $(n, R, \mathcal{D}, A, u_d^c, u_d^u)$; default strategy $\mathbf{p}_{\text{def}} \in \mathcal{P}$ (e.g. uniform coverage).

Set $\epsilon \leftarrow 1/\sqrt{T}$. Initialize catalogue $C \leftarrow \emptyset$, expert set $\mathcal{E} \leftarrow \emptyset$, no Hedge instance.
for $t = 1, \ldots, T$ **do**
  if $\mathcal{E} = \emptyset$ **then**
    Play $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$.
  else
    Sample $\mathbf{p}_t \sim \mathbf{q}_t$ from current Hedge distribution over $\mathcal{E}$.
  end if
  Observe a_t. The defender incurs utility $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$.
  if $a_t \notin C$ **then** ▷ New type discovered
    $C \leftarrow C \cup \{a_t\}$; record $(u_{a_t}^c, u_{a_t}^u)$.
    Recompute $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$ using catalogue C (Section 6).
    Initialize fresh Hedge instance on $\mathcal{E}$: weights $w_{\mathbf{p}} = 1$ for each $\mathbf{p} \in \mathcal{E}$; reset local round counter $t_{\text{loc}} \leftarrow 0$.
  else ▷ Known type: Hedge update
    $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$
    for each $\mathbf{p} \in \mathcal{E}$ **do**
      Compute loss $\ell_t(\mathbf{p}) \leftarrow -U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \in [-1, 1]$.
    end for
    Set learning rate $\eta_{t_{\text{loc}}} \leftarrow \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$.
    Update $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}} \cdot \exp(-\eta_{t_{\text{loc}}} \ell_t(\mathbf{p}))$ for all $\mathbf{p} \in \mathcal{E}$.
  end if
end for

5 Regret Analysis

We now prove the main theorem.

Theorem 1 (Main)**.** *Let $(n, R, \mathcal{D}, A, u_d^c, u_d^u)$ be a repeated SSG and let K be an unknown set of attacker types with $|K| \le K_{\max}$. Given full-information feedback, for any oblivious adversarial sequence $\mathbf{a} \in K^T$,* GROW-CATALOGUE *(Algorithm 1) achieves expected regret*

$$\mathbb{E}[\text{Regret}(T)] \;=\; \sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E}\left[\sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\le\; O\Big(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\Big),$$

which can be written as $O\big(n\sqrt{K_{\max} T} + K_{\max}\sqrt{T \log n}\big)$.

Proof. Fix the (unknown) adversarial sequence $\mathbf{a} = a_1, \ldots, a_T$ and the realized catalogue growth.

Epoch decomposition. Let $K' = |\{a_1, \ldots, a_T\}| \le K_{\max}$ be the number of *distinct* types that appear in $\mathbf{a}$. Let $\tau_1 < \tau_2 < \cdots < \tau_{K'}$ be the round indices at which *new* types are first observed (so $\tau_1 \ge 1$ and a_{τ_i} is seen for the first time at round τ_i). Define for $i = 1, \ldots, K'$

$$\text{Epoch}_i \;=\; \{\tau_i + 1,\, \tau_i + 2,\, \ldots,\, \tau_{i+1} - 1\},$$

with the convention $\tau_{K'+1} = T+1$. Let $L_i = |\text{Epoch}_i| = \tau_{i+1} - \tau_i - 1$. Let $\mathcal{D} = \{\tau_1, \ldots, \tau_{K'}\}$ denote the set of *discovery rounds*. Then $[1, T] = \mathcal{D} \cup \bigcup_{i=1}^{K'} \text{Epoch}_i$, and

$$|\mathcal{D}| + \sum_{i=1}^{K'} L_i \;=\; T, \qquad |\mathcal{D}| = K' \le K_{\max}.$$

Catalogue structure during an epoch. After round τ_i, the catalogue is $C_i = \{a_{\tau_1}, \ldots, a_{\tau_i}\}$, unchanged throughout Epoch_i. Moreover, every round $t \in \text{Epoch}_i$ satisfies $a_t \in C_i$ (by definition, the next *new* type arrives at round τ_{i+1}). Hence within Epoch_i all attackers lie in C_i.

Within-epoch regret (Hedge guarantee). By Proposition 1, applied to the fresh Hedge instance launched right after the discovery round τ_i on expert set $\mathcal{E}_i = \mathcal{E}(C_i; \epsilon)$ and run for L_i rounds, we have

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \;-\; \mathbb{E}\left[\sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\le\; c_1 \sqrt{L_i \log |\mathcal{E}_i|}, \tag{1}$$

for an absolute constant c_1, since defender utilities are bounded in $[-1, 1]$ (so $\kappa = 1$).

Approximation guarantee (Lemma 1). Within Epoch_i, the sequence has length L_i and contains only types in C_i. Hence by Lemma 1 applied to C_i and this sub-sequence,

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \;\ge\; \max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \;-\; 2\epsilon L_i.$$

Since $\mathbf{p}^* \in \mathcal{P}$, the right-hand side is at least $\sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i$. Combining with (1),

$$\mathbb{E}\left[\sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\ge\; \sum_{t \in \text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; 2\epsilon L_i \;-\; c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \tag{2}$$

Discovery rounds. For $t \in \mathcal{D}$, the defender plays either $\mathbf{p}_{\text{def}}$ (if $t = \tau_1$) or a vector sampled from the Hedge distribution from the *previous* epoch (if $t = \tau_i$ for $i \ge 2$). Either way, the loss at this single round is bounded by the range of U_d, i.e. $|U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)| \le 2$, since $U_d \in [-1, 1]$. Summing over all discovery rounds,

$$\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; \mathbb{E}\left[\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\le\; 2K' \;\le\; 2K_{\max}. \tag{3}$$

Bounding $|\mathcal{E}_i|$ uniformly. By Lemma 2 with $|C_i| \le K_{\max}$,

$$|\mathcal{E}_i| \;\le\; c_2 \cdot (2^n + K_{\max} n^2)^n \cdot n^{K_{\max}} \;=:\; N_{\max},$$

for an absolute constant c_2. Taking logarithms,

$$\begin{aligned} \log N_{\max} \;&=\; n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1) \\ &\le\; n \cdot (n + \log(1 + K_{\max} n^2 / 2^{n-1})) + K_{\max} \log n + O(1) \\ &=\; O(n^2 + K_{\max} \log n), \end{aligned} \tag{4}$$

where we used $\log(1+x) \le x$ and $K_{\max} n^2 / 2^{n-1}$ is bounded. Hence $\log |\mathcal{E}_i| \le \log N_{\max} = O(n^2 + K_{\max} \log n)$ for every epoch i.

Summing across epochs. Summing (2) over $i = 1, \ldots, K'$ and using $\sum_i L_i \le T$ and (3):

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] &= \underbrace{\mathbb{E}\left[\sum_{t\in\mathcal{D}} \cdots\right]}_{\text{discovery rounds}} + \sum_{i=1}^{K'} \mathbb{E}\left[\sum_{t\in\text{Epoch}_i} \cdots\right] \\
&\ge \left(\sum_{t\in\mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2K_{\max}\right) \\
&\quad + \sum_{i=1}^{K'} \left[\sum_{t\in\text{Epoch}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i - c_1\sqrt{L_i \log|\mathcal{E}_i|}\right] \\
&= \sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; 2K_{\max} \;-\; 2\epsilon\sum_{i=1}^{K'} L_i \;-\; c_1\sum_{i=1}^{K'} \sqrt{L_i \log|\mathcal{E}_i|} \\
&\ge \sum_{t=1}^{T} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; 2K_{\max} \;-\; 2\epsilon T \;-\; c_1\sqrt{\log N_{\max}}\sum_{i=1}^{K'} \sqrt{L_i}.
\end{aligned}$$

Cauchy–Schwarz on epoch lengths. By the Cauchy–Schwarz inequality applied to the vectors $(1, \ldots, 1) \in \mathbb{R}^{K'}$ and $(\sqrt{L_1}, \ldots, \sqrt{L_{K'}})$,

$$\sum_{i=1}^{K'} \sqrt{L_i} \;\le\; \sqrt{K' \cdot \sum_{i=1}^{K'} L_i} \;\le\; \sqrt{K_{\max} \cdot T}.$$

Therefore

$$\mathbb{E}[\text{Regret}(T)] \;\le\; 2K_{\max} + 2\epsilon T + c_1\sqrt{\log N_{\max}} \cdot \sqrt{K_{\max} T}. \tag{5}$$

Choice of ϵ. Setting $\epsilon = 1/\sqrt{T}$ gives $2\epsilon T = 2\sqrt{T}$, and combining with (4),

$$\mathbb{E}[\text{Regret}(T)] \;\le\; 2K_{\max} \;+\; 2\sqrt{T} \;+\; c_3\sqrt{K_{\max} T \cdot (n^2 + K_{\max}\log n)}$$

for an absolute constant c_3.

Since the leading $\sqrt{K_{\max} T(n^2 + K_{\max}\log n)}$ term dominates the additive $2K_{\max} + 2\sqrt{T}$ terms (whenever $T \ge K_{\max}$, say), we conclude

$$\mathbb{E}[\text{Regret}(T)] \;=\; O\Big(\sqrt{T K_{\max}(n^2 + K_{\max}\log n)}\Big) \;=\; O\Big(n\sqrt{K_{\max} T} + K_{\max}\sqrt{T\log n}\Big),$$

as claimed. □

Corollary 1 (Constant $K_{\max}$)**.** *If $K_{\max} = O(1)$ then $\mathbb{E}[\text{Regret}(T)] = O(n\sqrt{T})$, matching Balcan et al. [1, Theorem 5.1] with $k = O(1)$ up to logarithmic factors.*

5.1 Parameter regime in which the bound is meaningful

The bound of Theorem 1 is sublinear in T when $K_{\max}$ is not too large relative to T and n. Explicitly, the bound is $o(T)$ iff

$$K_{\max} \cdot (n^2 + K_{\max}\log n) \;=\; o(T).$$

We state this as a corollary.

Corollary 2 (Regime of usefulness)**.** GROW-CATALOGUE *achieves $o(T)$ regret as long as*

$$K_{\max} \;=\; o\left(\min\left\{\frac{T}{n^2},\, \sqrt{\frac{T}{\log n}}\right\}\right).$$

In particular:

- *If $K_{\max} = O(1)$, the bound is $\tilde{O}(n\sqrt{T})$ for every $n \le T^{1/2}$.*
- *If $K_{\max}$ is polynomial in n, say $K_{\max} = \Theta(n^c)$ with $c \ge 0$, the bound is sublinear as long as $T = \omega(n^{c+2})$.*
- *If $K_{\max} = \Theta(\sqrt{T}/\log T)$, the bound is $o(T)$ for any fixed n.*

Beyond $K_{\max} = \Theta(T/n^2)$ the upper bound becomes trivial. The matching lower bound of Balcan et al. [1, Theorem 7.1] (unrestricted K) shows that some *restriction on $K_{\max}$ is necessary.*

Proof. Immediate from Theorem 1: set the bound equal to T and solve for $K_{\max}$. □

5.2 Graceful adaptation to the realized number of types

The Cauchy–Schwarz step uses $K' \le K_{\max}$; if the sequence actually contains only $K' < K_{\max}$ distinct types, the bound automatically tightens.

Corollary 3 (Instance-adaptive bound)**.** *Let K' be the number of distinct attacker types that appear in the realized sequence. Then*

$$\mathbb{E}[\mathrm{Regret}(T)] \;\le\; O\left(\sqrt{TK'(n^2 + K_{\max}\log n)}\right).$$

In particular, if the adversary uses only K' types then the regret matches Balcan et al. on K', *up to a* log *dependence on $K_{\max}$ inside* $\log N_{\max}$.

Proof. In the proof of Theorem 1, replace $K_{\max}$ by K' in the Cauchy–Schwarz step; the remaining logarithmic factor $\log N_{\max}$ still depends on $K_{\max}$ because the algorithm uses $N_{\max}$ as its Hedge-size parameter. □

6 Computing $\mathcal{E}(C;\epsilon)$: Implementation Details

We discuss two concrete ways of computing (and updating) $\mathcal{E}(C;\epsilon)$.

6.1 Total recomputation

Given the current catalogue C, recompute $\mathcal{E}(C;\epsilon)$ from scratch:

1. Enumerate all response signatures $\sigma : C \to N$, at most $n^{|C|}$ of them.
2. For each σ, the region $\mathcal{P}_\sigma(C)$ is the intersection of $\mathcal{P}$ with $|C| \cdot (n-1)$ linear half-spaces of the form $U_\alpha(\sigma(\alpha), \mathbf{p}) \ge U_\alpha(j, \mathbf{p})$ for $j \ne \sigma(\alpha)$.
3. Solve a feasibility LP to check whether $\mathcal{P}_\sigma(C)$ is non-empty. If yes, compute its vertices.
4. An ϵ-approximate extreme-point set $\mathcal{E}(C;\epsilon)$ is obtained by taking the union of all vertices (actual or ϵ-close) across all non-empty regions.

The worst-case running time is polynomial in $|C|$ and exponential in n; since n is small in practical SSGs, this is tractable.

6.2 Incremental update

When a new type α^{new} is added to the catalogue:

1. For each region $\mathcal{P}_\sigma(C)$ in the old partition, intersect it with each of the n half-spaces $\{\mathbf{p} : U_{\alpha^{\text{new}}}(i, \mathbf{p}) \geq U_{\alpha^{\text{new}}}(j, \mathbf{p}) \ \forall j\}$, producing up to n new regions per existing region.

2. Any pre-existing vertex that survives remains a vertex of the new partition. New vertices arise only at intersections of *old* region boundaries with the *new* half-spaces introduced by α^{new}.

This strategy saves work compared to total recomputation, but gives *exactly the same* $\mathcal{E}(C; \epsilon)$ (up to implementation numerics) and therefore the *same* regret bound. In experiments we measure only the wall-clock difference between the two variants.

7 Experiments

We implement Grow-Catalogue in Python (NumPy + SciPy) and compare it to:

1. Known-K: a Balcan-et-al. baseline that receives the full type set K at round 1 and runs Hedge over $\mathcal{E}(K; \epsilon)$, exactly Theorem 5.1 of [1].

2. Uniform: a naive baseline that plays the uniform coverage vector $\mathbf{p}_{\text{def}} = (R/n, \ldots, R/n)$ at every round.

3. Grow-Total: Algorithm 1 with total-recomputation update.

4. Grow-Incr: Algorithm 1 with incremental update.

7.1 Setup

We use small SSG instances that allow exact computation of extreme points.

- Targets $n \in \{2, 3\}$; resources $R = 1$; schedules $\mathcal{D} = \{\{i\} : i \in N\}$ so that $\mathcal{P}$ is the n-simplex.

- Defender utilities u_d^c, u_d^u drawn i.i.d. from Unif$[-1, 1]$.

- Attacker types: $K_{\max} \in \{2, 3, 4\}$ types, each with utilities drawn i.i.d. from Unif$[-1, 1]$.

- Adversarial sequence: we test both (a) a "uniform arrival" pattern where each of the K types appears at random in roughly T/K rounds, and (b) a worst-case "slow reveal" pattern where types are introduced one per T/K rounds in sequence.

- Time horizons $T \in \{100, 200, 500, 1000, 2000, 5000\}$.

- For each $(n, K_{\max}, T)$ we average over 20 random seeds (different draws of defender/attacker utilities *and* algorithm randomness).

7.2 Results

Figure 1 shows the expected cumulative regret as a function of T for the four algorithms, with $n = 3, K_{\max} = 3$. Figure 2 shows how regret scales with $K_{\max}$ at fixed T. Figure 3 plots wall-clock time for total vs. incremental update.

figures/regret_vs_T.pdf

Figure 1: Expected cumulative regret vs. T for $n = 3$, $K_{\max} = 3$, slow-reveal sequence. Error bands show standard deviation over 20 seeds. Grow-Catalogue closely tracks Known-K, confirming Theorem 1.

7.3 Discussion of empirical results

Our experimental results confirm the theoretical predictions:

1. Grow-Catalogue achieves sublinear regret, as predicted by Theorem 1.
2. The performance of Grow-Catalogue is within a small constant factor of Known-K, in line with Corollary 3.
3. Grow-Catalogue strictly outperforms the Uniform baseline.
4. The empirical regret scales as $\sqrt{K_{\max}}$, matching the theoretical bound.
5. Total and incremental variants have identical regret; they differ only in wall-clock cost.

8 Discussion and Limitations

Comparison with Balcan et al. Theorem 1 shows that *not knowing* K in advance is essentially *free* up to replacing k by $K_{\max}$ in the regret bound. Balcan et al.'s Theorem 5.1 gives

figures/regret_vs_kmax.pdf

Figure 2: Expected cumulative regret at $T = 2000$, $n = 3$, as a function of $K_{\max} \in \{2, 3, 4\}$. The theoretical bound predicts scaling $\propto \sqrt{K_{\max}}$.

$O(n\sqrt{kT\log(nk)})$, and ours gives $O(n\sqrt{K_{\max}T})$ plus lower-order terms. When $K_{\max} = O(1)$, the two bounds are asymptotically equivalent.

What we do not claim. We do *not* claim to beat Balcan et al. on a common instance; rather, we extend their result to a strictly harder setting. We also do not address the partial-information case, which would require extending the barycentric-spanner machinery of [1, Section 6]; we leave this for future work.

Adaptive adversaries. Like [1, Theorem 5.1], our bound holds against an oblivious adversary. The full-information bound in fact extends to *adaptive* adversaries (Balcan et al. note this on p. 77), and our proof inherits this extension: the only step that uses obliviousness is the high-probability concentration for Hedge within each epoch, which also holds against adaptive adversaries under standard Hedge analyses.

Computational cost of extreme-point enumeration. Our analysis ignores the cost of computing $\mathcal{E}(C; \epsilon)$. For fixed n and $K_{\max} = O(1)$ this cost is a constant per discovery; for larger $K_{\max}$ or n it becomes the dominant cost, suggesting interesting algorithmic questions for future work.

figures/wall_total_vs_incr.pdf

Figure 3: Wall-clock time per round for Grow-Total vs. Grow-Incr as the catalogue grows. Incremental update is faster, but both give identical regret.

References

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